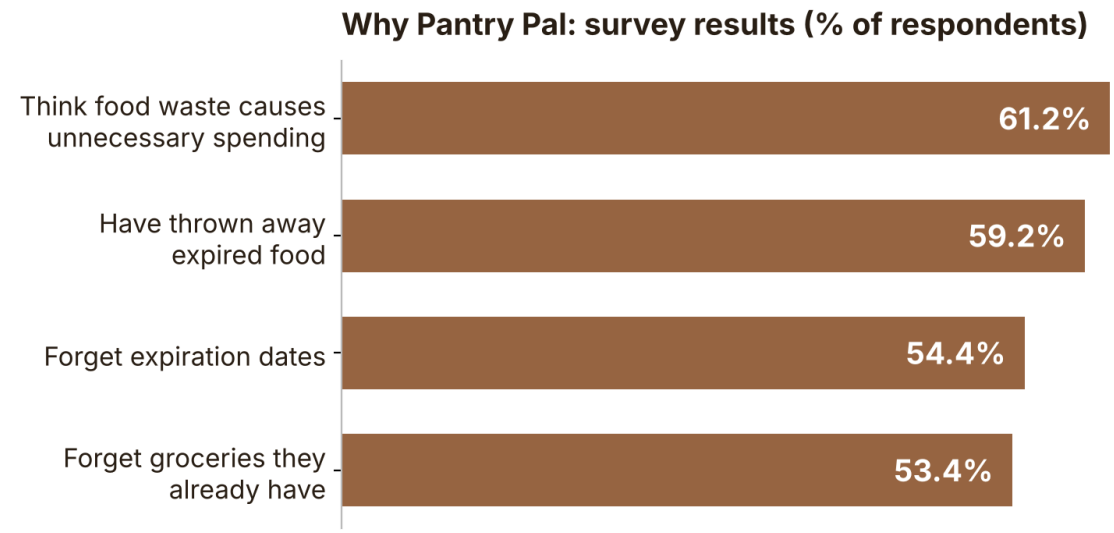
Pantry Pal – Final Report

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1. Introduction and Problem

Pantry Pal is a grocery expiration tracking app for parents, working adults and anyone who stores groceries at home. People forget what they own and when it expires, so food is wasted, money is lost, and expired food may be eaten. Our survey shows how common this is.



Source: Pantry Pal team survey (demo slides)

2. Objectives and Scope

The app helps users organise groceries, track expiry dates, get reminders, see dates in a list and a calendar, and (Premium) review used versus wasted food, to reduce household food waste. Out of scope: online shopping, delivery, nutrition tracking, spoilage detection, supermarket integration and messaging.

3. Requirements

The SRS (v1.0.1) defines 11 functional requirements (FR), each a user story with testable acceptance criteria, traced to a problem and a test case (TC-01 to TC-11) in the Golden Thread matrix.

FR	Feature	Plan
FR-1	Account management and Premium upgrade	Free
FR-2	Add, edit, delete and mark items as used	Free
FR-3	Automatic expiry estimate from category and purchase date	Free
FR-4	Expiration reminders (default 3 days)	Free
FR-5	Active items list	Free
FR-6	Expiring and expired items	Free
FR-7	Receipt and item scanning with a review list	Premium
FR-8	Automatic category detection	Premium
FR-9	Custom reminder period (1 to 30 days)	Premium
FR-10	Expired history (7 days Free, unlimited Premium)	Free / Premium

FR-11	Usage summary: daily, weekly, monthly	Premium

The 11 non-functional requirements are also testable, for example saving data within 2 seconds, a first-time user adding an item within 60 seconds, and 99% availability.

4. Design and Prototype

Design artefacts are a use-case diagram (Household Member as the actor, 14 use cases) and a sequence diagram covering login, add grocery, view and edit, reminders, mark as used and the usage summary.

The final prototype is a clickable, responsive web app with Pantry, Calendar, Notifications and Profile screens. Key rules: expiry is estimated from a category shelf life (for example 3 days for meat, 7 for dairy, 90 for frozen) and stays editable; items show Expiring Soon within 3 days; marking an item as used asks how much, keeps the rest active and offers undo. Premium (simulated, $1.99 per month) adds receipt scanning with a review list, custom reminders and the usage summary, which resets monthly. Passwords need 8+ characters and are stored as salted SHA-256 hashes.

5. Process and Technologies

We followed Agile in short iterations: charter (M1), SRS (M2), design, prototype (M3), then testing. The prototype grew from v1.0 (barcode scanning, three tabs, plain-text passwords) to the final version (receipt scanning, calendar, Notifications screen, hashed passwords). Tools: Readdy for the prototype screens, HTML, CSS and JavaScript, browser local storage for data, Hatchable and GitHub Pages for hosting, and GitHub for version control.

6. Use of AI

We used ChatGPT, Claude, Gemini, Perplexity and Readdy AI, recorded in a dated AI usage log. For requirements, AI suggested FRs, NFRs and use cases; we changed the actor to Household Member, fixed FR and use-case links and cut unnecessary detail. AI also checked our files against the FRs. For the prototype, AI suggested layouts, HTML, CSS and JavaScript and explained errors; we tested everything, rewrote code that failed and reviewed it together.

7. Limitations, Future Work and Conclusion

This is a prototype: data is stored on the device only, so there is no sync and the per-user security requirement is only partly met; receipt scanning needs a server with an AI key, otherwise sample items are shown; the Premium upgrade takes no payment; and TC-01 to TC-11 results are not recorded here. Next steps are cloud accounts (Supabase or Readdy Backend), AI receipt extraction and timed usability tests.

All 11 requirements trace from a stated problem to a built feature, giving a working base for real households.